

# MATTHEW G. JOHNSTON, PhD

Hands-on AI engineer and CTO of a 100-store PE-backed retail chain on the exec board with P&L oversight. I deliver agent-ready systems as part of value creation, while running traditional data science, engineering and IT teams. I value working in environments where we can make significant change quickly, especially using AI systems as a value multiplier to reduce manual task and free up human time for valuable work.

## EXPERIENCE

Sep 2024 – Present

Chief Technology Officer, *Jollyes Pets, TDR Portfolio Company*

Operating Board member with P&L oversight; lead three functions - IT (multi-site retail across 100+ stores), data engineering, data science. Co-authored group strategy.

**Agentic analytics platform** · Designed and shipped a curriculum-gated MCP server (about → describe\_table → query) giving every manager natural-language access to data. Enforces context tiering, self-corrects on misuse, runs ~\$10/month, always-on.

**Production personalisation** · Compressed 600k loyalty members into 20-character in-memory fingerprints – the same product powers real-time recommendations (<1ms) and voucher audience targeting

**Customer-facing technical delivery at scale** · IT function deploys, integrates, and supports technology in-person across 100+ retail sites with non-technical end-users

**Hands-on engineering** · Payments middleware (Node.js / Postgres / AWS / Terraform) shipped in 6 weeks: -99% failed payments. PCI-DSS scope owned end-to-end.

**Commercial outcomes** · Launched Q-com (last-mile) across 100+ stores end-to-end - commercials, ops, IT, marketing: +2% LFL in 6 weeks; 40% of group growth. LLM-embeddings + XGBoost store-turnover model now gates board sign-off for new sites.

Aug 2023 – Sep 2024

Head of Data Science (Marketing), *Sainsbury's*

Ran two DS teams (12 ICs) and an 11-person engineering team on the Nectar dataset (20m customers)

Owned end-to-end delivery of on-site recommendations - material revenue driver (>£100m) cited in quarterly results. Translated segmentations into platform priorities for the central Insights team. Won DataIQ "AI and data for good" hero organisation award (2024) for helping customers make healthier choices.

May 2022 – Aug 2023

Data Science Manager, *Sainsbury's*

Two consecutive teams: personalised pricing and Nectar segmentation. Launched a personalised offer mechanic delivering £10m incremental revenue in 6 weeks. Tripled Argos's Meta ROAS.

Sainsbury's Employee of the Year · Sainsbury's Future Leader · DataIQ Future Leader 2024

Jul 2021 – Apr 2022

Senior Data Consultant, *Artefact*

Led a 10-person delivery team (consultants, scientists, engineers) embedded into a global FMCG client - direct customer-site delivery; 1P data activations and global measurement framework.

## SELECTED TECHNICAL WRITING

"A stateful MCP analyst: enforced agent curriculum and attention management" (Apr 2026) - production MCP server design notes.

"Fingerprinting 600,000 customers in 20 characters each" (May 2026)

"Embeddings: LLM's best kept secret?" (Nov 2025) - combining LLM embeddings with traditional ML

## EDUCATION

Executive

The Cambridge and Financial Times Strategy Programme, *Cambridge Judge Business School*

PhD

Biology and Statistics PhD, Composition of and interactions at plasmodesmata, *John Innes Centre Publications* · Peer-reviewed inc. Science, PNAS, Science Advances, Journal of Experimental Botany

MA

Class I, Natural Sciences, MA, *Christ's College, University of Cambridge*